Fiscal Year:	FY 2018	**Task Last Updated:**	FY 06/11/2018
PI Name:	Hirsa, Amir Ph.D.		
Project Title:	Amyloid Fibril Formation in Microgravity: Distinguishing Interfacial and Flow Effects		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	09/01/2013	**End Date:**	09/30/2020
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	5	**No. of Master' Degrees:**	3
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	15	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 9/30/2020 per PI and NSSC information (Ed., 6/4/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lopez, Juan Ph.D. (Arizona State University)		
Grant/Contract No.:	NNX13AQ22G		
Performance Goal No.:			
Performance Goal Text:			

Formation of amyloid fibrils is widely studied because of their role in neurodegenerative diseases, including Alzheimer's. The fibrillization process is characterized by a nucleation step followed by fibril growth, much like the steps that occur during protein crystallization. Aside from the general interest in macromolecular biophysics, recent evidence shows the relevance of amyloid fibril formation in animal models exposed to cosmic radiation, with obvious implications for interplanetary travel. A preponderance of evidence shows that agitating protein solutions can significantly accelerate amyloid fibril formation; however, the mechanisms by which the presence of bubbles and flow independently affect fibrillization are not known. The situation is further complicated because we do not yet have mechanistic models that can quantify the influence of natural convection due to gravity on the fibrillization process. We know that fluid interfaces (e.g., air-water interface) induce amyloid fibrils when they are sheared, but the influence of

Task Description:	container walls is not fully understood. Here, we aim to distinguish the influence of flow and fluid interfaces on amyloid formation by removing other effects, including solid walls and buoyancy effects. Formation of protein amyloid fibrils provides a model system for studying biomolecular transport phenomena, where the influence of flow can be altogether negated by the elimination of gravity. Furthermore, the experiments can be performed aboard the International Space Station (ISS) with essentially no solid container, since surface tension can provide liquid containment. Thus, the free surface can be readily observed (e.g., with high magnification objectives), manipulated (e.g., can be sheared or left otherwise undisturbed), and modified (e.g., by spreading a monomolecular oil film). First, we will perform experiments at 1g on a small droplet to ascertain the effect of forced convection on a liquid surface. Next, we will generalize our very successful computational model in annular systems to the spherical geometry of liquids at microgravity. We will design the space-based experiments based on the results from the ground experiments and numerical predictions. Finally, we will perform experiments aboard ISS on large-scale drops in an experiment referred to as the Ring-Sheared Drop (RSD).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	When proteins are perturbed from their native state, they lose their native confirmation (folding) and then can bond with other proteins, forming an aggregate. This is the reason why the transparent fluid in an egg becomes a white solid when cooked. Amyloid fibrils are a unique type of protein aggregate that occur naturally in the body. We study them because amyloid plaques in the brain are associated with many neurodegenerative disorders, including Alzheimer's. Fibrils are very intriguing because they are highly ordered and self-assemble. When heated, insulin molecules, for example, denature and bond with other denatured molecules (or assists in denaturing the other molecules). These form a nucleation site, and from there the fibril elongates as more molecule bond into what is called a "beta sheet" structure. The primary mechanism for activation of protein in the body is not yet understood, but it is likely to be a combination of flow friction, aka "shear stress," and interfacial effects that native proteins are exposed to. Aside from their central role in neurodegenerative diseases such as Parkinson's and Alzheimer's, formation of amyloid structures in proteins plays a role in other diseases including diabetes.
Task Progress:	The following is a list of accomplishments made during Year-5 of the NASA project NNX13AQ22G. I) Progress in Administration A) The SRD (Science Requirements Document), which was initiated in Year-1 and revised in Years 2, 3, and 4 was finalized and signed in Year-5. Many of the changes to the requirements reflect practical limitations and other considerations revealed during conferences with other scientists and engineers at Marshall Space Flight Center (MSFC) and their prime contractor for the RSD (Teledyne-Brown Engineering). B) The PDR (Preliminary Design Review) was held in Huntsville AL on August 11, 2017 and successfully passed. C) The CDR (Critical Design Review) was held in Huntsville AL on February 13, 2018 and successfully passed. D) A second parabolic flight campaign titled "Protein-Drop Pinning in Microgravity" was successfully undertaken in support of the RSD project. The project was funded by a separate grant (NNX17AG04G) through the NASA Flight Opportunities program. The two-day flight campaign took place on Nov. 14th and 15th, 2017. II) Scientific and Technological Progress 1) Extensive computational studies of the flow field in the RSD geometry were conducted and the results are available online in the International Journal of Heat and Mass Transfer (Gulati et al.) and are due for print publication in October 2018 (See Bibliography section). In the study, the mixing characteristics inside the geometry of the RSD were examined in detail. As the first journal paper on a spherical system where the flow is driven by the action of surface shear viscosity, we examined the limit of large Boussinesq number, where the effect of surface shear viscosity dominates over the viscosity in the bulk. 2) The restriction of infinite Boussinesq number has been lifted in a recent study in the flow in the RSD. Preliminary results show an unexpected behavior. Additional studies are being completed at the time of this writing. Once complete, the results will be submitted for publication in Physical Review Fluids. 3) Extensive experimental studies are being undertaken on amyloid fibril formation and interaction with solid boundaries. Novel patterns of amyloid fibrils were observed to form with a striking correction to the shear normal to the interface. Once completed, the results will be submitted for publication in an appropriate journal. 4) A combined theoretical, computational, and experimental study of the coupling between a non-Newtonian interface and a Newtonian bulk fluid was conducted and the results have been submitted for publication in the Physical Review Letters (Raghunandan et al., Predicting shear rheology of condensed-phase monomolecular films at the air-water interface). The results showed that for a model system, monolayers of the phospholipid DPPC, the primary constituent of lung surfactant and the membrane that forms our cell walls, the response to shear can be accurately predicted over 6 decades of shear rate and more than two decades of Reynolds number. Note that DPPC was utilized in this first phase of the non-Newtonian modeling effort since it is already well characterized by several groups. The modeling effort for amyloids is completely new and now that we have the machinery for DPPC working, we are in a position to tackle the challenge of predicting the shear response of interfaces with amyloid fibrils. 5) An extensive experimental study has been underway to examine the fibrillization kinetics of insulin in a cylindrical analog of the RSD in the laboratory, namely the knife-edge surface viscometer. After the results are reproduced, they will be sent for publication in a suitable journal. 6) An experimental apparatus has been designed and assembled to study the effects of oxygen on shear-induced fibrillization on the surface of protein solution. The flow apparatus utilizes a deep-channel viscometer geometry. An enclosure surrounding the flow apparatus ensures that only an inert gas (Argon in this case) comes in contact with the protein solution. 7) A two-day parabolic flight campaign was undertaken aboard ZeroG Corp. aircraft in support of technology development for design of the RSD aboard the ISS. We successfully demonstrated the ability of a simple strategy to pin the drop with a diameter of 1.5 cm, which is about 60% of the diameter of the RSD. Liquids that were successfully

	pinned include DI water, DI water with fluorescein dye (10 ppm), standard growth media for the cultivation of cyanobacteria, and protein solutions with 2 mg/mL of insulin. In several parabolas, drops were pinned despite a significant amount of g-jitter caused by clear-air turbulence. The data collected during those flights is still being analyzed. The results will be subsequently published.
Bibliography Type:	Description: (Last Updated: 04/15/2026)
Abstracts for Journals and Proceedings	Raghunandan A, Underhill PT, Lopez JM, Hirsa AH. "Predicting nonlinear shear rheology of soft interfaces." Presented at the 92nd ACS Colloid & Surface Science Symposium, State College, PA, June 10-13, 2018. 92nd ACS Colloid & Surface Science Symposium, State College, PA, June 10-13, 2018. , Jun-2018
Abstracts for Journals and Proceedings	Gulati S, Riley FP, Hirsa AH, Lopez JM. "Effects of finite surface shear viscosity in spherical systems." Presented at the 92nd ACS Colloid & Surface Science Symposium, State College, PA, June 10-13, 2018. 92nd ACS Colloid & Surface Science Symposium, State College, PA, June 10-13, 2018. , Jun-2018
Abstracts for Journals and Proceedings	Riley F, Gulati S, Hirsa AH, Lopez JM. "Mixing in millimeter-scale drops by the action of surface viscosity." 70th Annual Meeting of the APS Division of Fluid Dynamics, Denver, CO, November 19-21, 2017. Bulletin of the American Physical Society. 2017 Nov;62(14): BAPS.2017.DFD.A15.6. http://meetings.aps.org/link/BAPS.2017.DFD.A15.6 , Nov-2017
Abstracts for Journals and Proceedings	Lopez JM, Raghunandan A, Underhill PT, Hirsa AH. "Modeling and measuring non-Newtonian shear flows of soft interfaces." 70th Annual Meeting of the APS Division of Fluid Dynamics, Denver, CO, November 19-21, 2017. Bulletin of the American Physical Society. 2017 Nov;62(14):BAPS.2017.DFD.E38.6. http://meetings.aps.org/link/BAPS.2017.DFD.E38.6 , Nov-2017
Abstracts for Journals and Proceedings	Raghunandan A, Gulati S, Riley FP, Hirsa AH, Lopez JM. "Ring-Sheared Drop (RSD) module for the study of amyloid fibril formation." Presentation at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Articles in Peer-reviewed Journals	Gulati S, Riley FP, Lopez JM, Hirsa AH. "Mixing within drops via surface shear viscosity." International Journal of Heat and Mass Transfer. 2018 Oct;125:559-68. Epub 2018 Apr 27. https://doi.org/10.1016/j.ijheatmasstransfer.2018.04.057 , Oct-2018